SYSTEM DOCUMENTATION & TECHNICAL REPORT

Alzheimer's Disease Classification

Medical Disclaimer: This classification system is built strictly for educational and demonstration purposes. Model outputs do not constitute a clinical diagnosis, risk assessment, or medical decision support.

1. Executive Summary & Objective

This project implements an end-to-end Machine Learning pipeline designed to classify patient records regarding potential Alzheimer's disease diagnosis. The system analyzes multi-domain features encompassing demographics, lifestyle factors, clinical history, cognitive evaluations, behavioral markers, and functional assessment metrics.

Pipeline Scope

- **Data Processing:** Quality audits, missing value checks, and normalization.
- **Feature Engineering:** Risk aggregates, hemodynamics, and lipid indices.
- **Modeling:** Baseline benchmarks & cross-validated hyperparameter tuning.
- **Deployment:** Model serialization and Streamlit web interface.

Production Configuration

The production inference engine uses a tuned **Gradient Boosting Classifier** delivering robust predictive power across imbalanced class distributions.

Serialized Artifact: `final_gradient_boosting.pkl`

2. Dataset & Target Specification

The system was developed and validated on structured clinical observations stored in `alzheimers_disease_data(1).xlsx` (Sheet: `alzheimers_disease_data`).

2,149	36	64.6%	35.4%
TOTAL INSTANCES	FINAL FEATURES	CLASS 0 RATIO	CLASS 1 RATIO

Target Class Distribution

Diagnosis Class	Clinical Meaning	Count	Proportion
0	No Alzheimer's Disease	1,389	64.63%
1	Alzheimer's Disease Diagnosis	760	35.37%

*Due to class imbalance, accuracy was secondary; macro-averaged F1-Score and ROC-AUC were primary evaluation metrics.

3. Exploratory Data Analysis & Preprocessing

Data Quality & Distribution Summary

- **Missing Data:** 0 missing values detected across all 35 initial variables.
- **Outlier Audit:** 1.5 × IQR boundary checks revealed zero extreme anomalies requiring removal.
- **Skewness Evaluation:** Numerical features exhibited skewness between -0.07 and +0.05 (near-symmetric), removing the need for log transformations.

Core Feature Profile

Domain	Feature	Mean	Median	Range (Min – Max)
Demographics	Age	74.91	75.00	60 – 90
	BMI	27.66	27.82	15.01 – 39.99
Lifestyle	Physical Activity	4.92	4.77	0.00 – 9.99
	Diet Quality	4.99	5.08	0.00 – 10.00
	Sleep Quality	7.05	7.12	4.00 – 10.00
Hemodynamics	Systolic BP	134.26	134.00	90 – 179
	Diastolic BP	89.85	91.00	60 – 119
Lipid Profile	Cholesterol Total	225.20	225.09	150.09 – 299.99
	Cholesterol HDL	59.46	59.77	20.00 – 99.98
Assessment	MMSE	14.76	14.44	0.00 – 29.99
	Functional Assessment	5.08	5.09	0.00 – 10.00
	ADL (Activities of Daily Living)	4.98	5.04	0.00 – 10.00

4. Feature Engineering & Selection

Four engineered features were synthesized to capture complex multi-system interactions:

1. CognitiveSymptomCount (Range: 0–7)

$$MemoryComplaints + BehavioralProblems + Confusion + Disorientation + PersonalityChanges + DifficultyCompletingTasks + Forgetfulness$$

2. CardiometabolicRiskCount (Range: 0–4)

$$Smoking + CardiovascularDisease + Diabetes + Hypertension$$

3. PulsePressure & Total_HDL_Ratio

$$PulsePressure = SystolicBP - DiastolicBP \quad | \quad Total_HDL_Ratio = CholesterolTotal / CholesterolHDL$$

Pruning Strategy: Administrative identifiers `PatientID` and `DoctorInCharge` were dropped to prevent leakage and spatial confounding.

5. Preprocessing & Architecture

Data was partitioned into an 80% training set (1,719 rows) and a 20% stratified test set (430 rows) with `random_state=42`.

- **Continuous Features:** Scaled using `StandardScaler`.
- **Categorical Features:** `Ethnicity` and `EducationLevel` encoded using `OneHotEncoder(handle_unknown="ignore", drop="first")`.
- **Binary Features:** Passed through without alteration.

6. Model Benchmarking & Hyperparameter Tuning

Four classification algorithms were benchmarked on the preprocessed training set using 5-Fold Stratified Cross-Validation.

Baseline Evaluation Results

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Gradient Boosting (Baseline)	**94.42%**	**92.11%**	**92.11%**	**92.11%**	**93.89%**
Decision Tree	94.42%	92.11%	92.11%	92.11%	93.89%
Random Forest	93.49%	94.29%	86.84%	90.41%	91.98%
Logistic Regression	82.33%	75.68%	73.68%	74.67%	80.37%

Hyperparameter Optimization

`GridSearchCV` was executed on the Gradient Boosting baseline targeting the F1 metric:

```
Best Parameters (Gradient Boosting): { 'learning_rate': 0.05, 'max_depth': 2, 'min_samples_leaf': 5, 'min_samples_split': 2, 'n_estimators': 150 }
```

7. Production Model Performance

Evaluating the optimized Gradient Boosting Classifier on the 430 held-out test records:

95.58%	94.63%	92.76%	93.69%
ACCURACY	PRECISION	RECALL	F1 SCORE

Generalization Gap

- **Train Accuracy / F1:** 95.81% / 93.99%
- **Test Accuracy / F1:** 95.58% / 93.69%
- **Variance Gap:** ~0.23% (No significant overfitting)

Confusion Matrix Breakdown

	Pred 0	Pred 1
Actual 0	**270** (TN)	**8** (FP)
Actual 1	**11** (FN)	**141** (TP)

8. Deployment Architecture & Environment

Inference Execution Flow

> User Input Form ➔ Compute Derived Features ➔ 36-Feature Vector ➔ Preprocessor (.transform()) ➔ Model Prediction (.predict_proba()) ➔ Web UI Display

Project Repository Topology

```
Alzheimers-Disease-Classification/ ├── README.md ├── app.py # Streamlit application UI ├── requirements.txt # Environment specifications ├── final_gradient_boosting.pkl # Serialized production model ├── preprocessor.pkl # Serialized Transformer ├── deployment_info.pkl # Deployment metadata ├── model_summary.pkl # Model performance summary ├── notebooks/ | └── alzheimers_disease_classification.ipynb └── data/ └── alzheimers_disease_data(1).xlsx
```

Execution Commands

```
# Initialize virtual environment python -m venv venv venv\Scripts\activate # On Windows # Install locked dependencies pip install -r requirements.txt # Launch interactive Streamlit interface streamlit run app.py
```

9. Environment Versioning

Library / Dependency	Deployment Version	Purpose
`python`	3.13.9	Core Runtime Environment
`scikit-learn`	1.7.2	Model & Transformer Pipeline
`numpy`	2.2.6	Numerical Computations
`joblib`	1.5.2	Artifact Serialization
`streamlit`	Latest	Web Application Framework